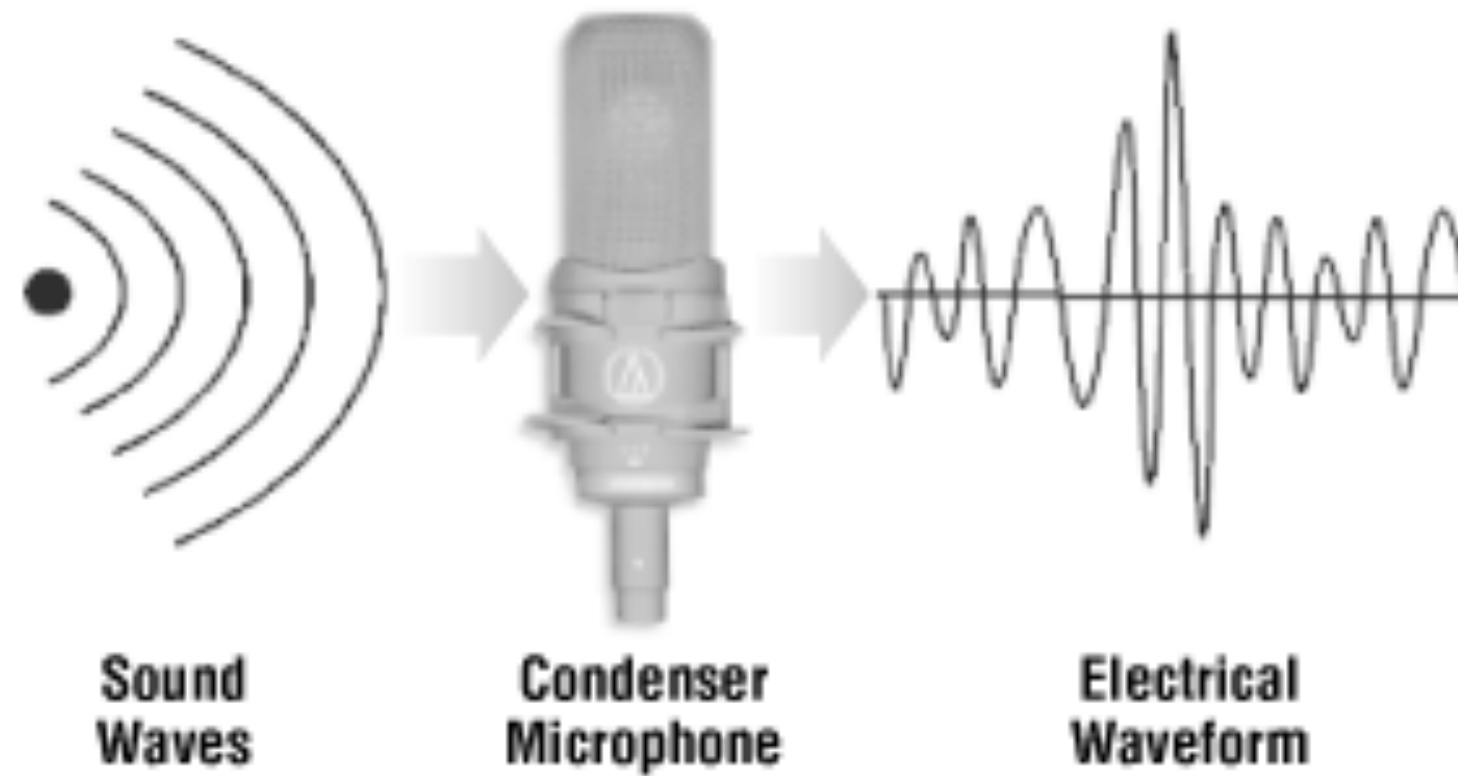


Microphones

What is a microphone?

- A pickup element to convert sound pressure into electrical voltages mounted in a box.

Sound to electricity



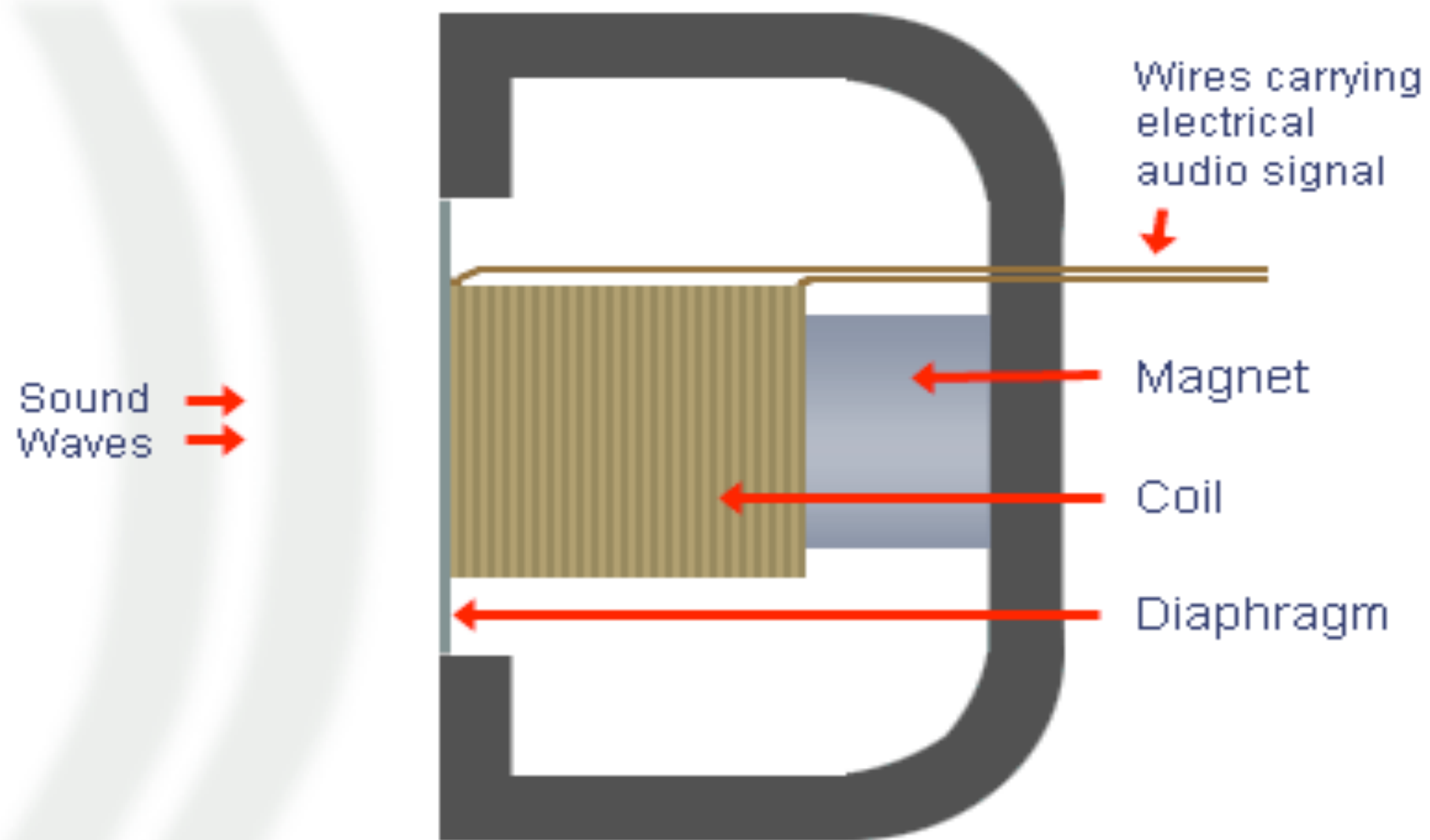
The two factors to consider when choosing a microphone are:

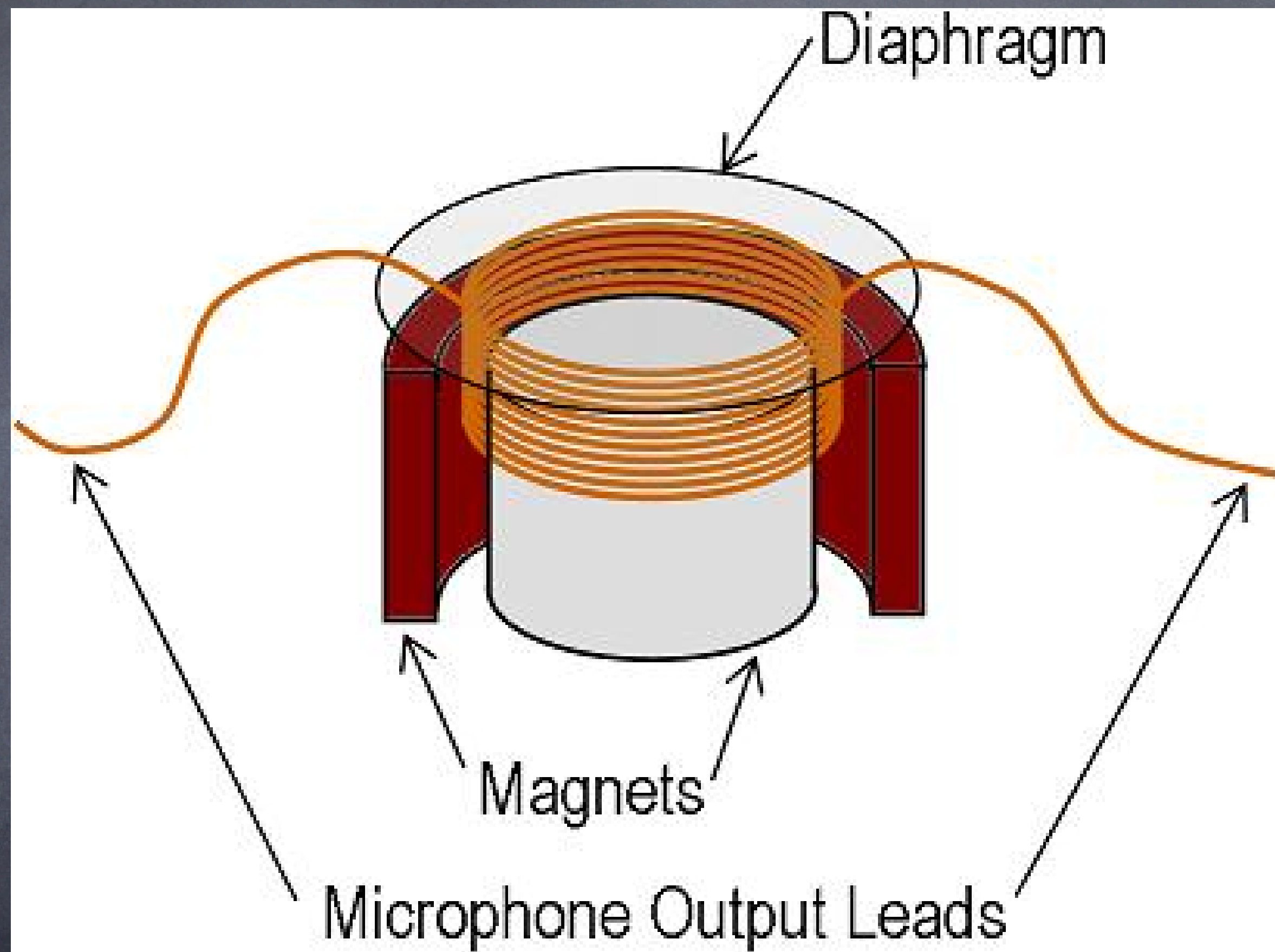
- 1. Its transducer design, or the method of converting sound waves into electrical waveforms
- 2. Its directionality, or pick-up pattern of surrounding sound

Transducer designs – Dynamic mics

- ⑥ A **dynamic** mic works like a speaker in reverse.
- ⑥ A coil of wire is mounted on the back of the diaphragm and located in the magnetic field provided by a permanent magnet.
- ⑥ When sound waves push the diaphragm, an electric signal is generated on the coil as it moves in the magnetic field.

Cross-Section of Dynamic Microphone

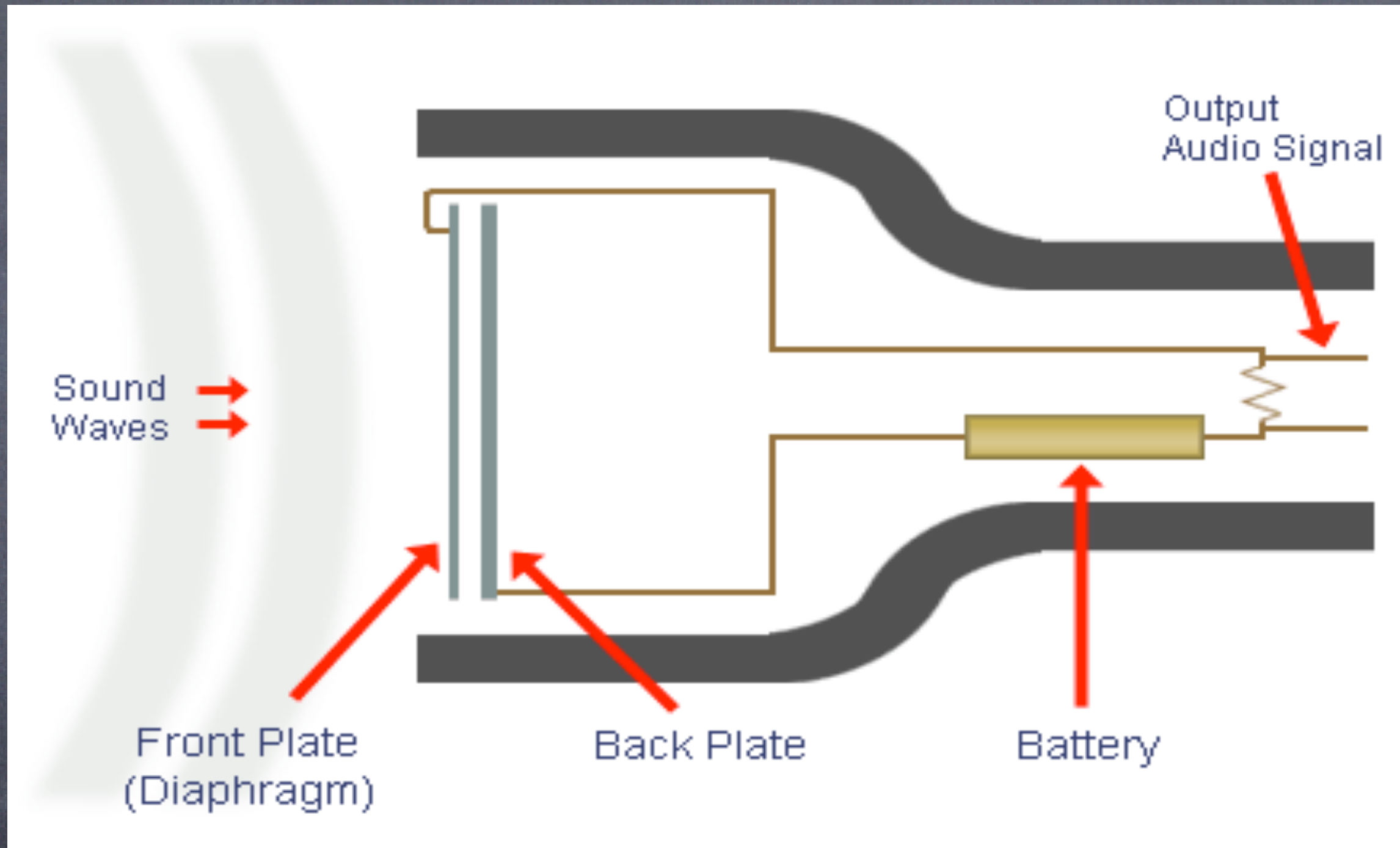




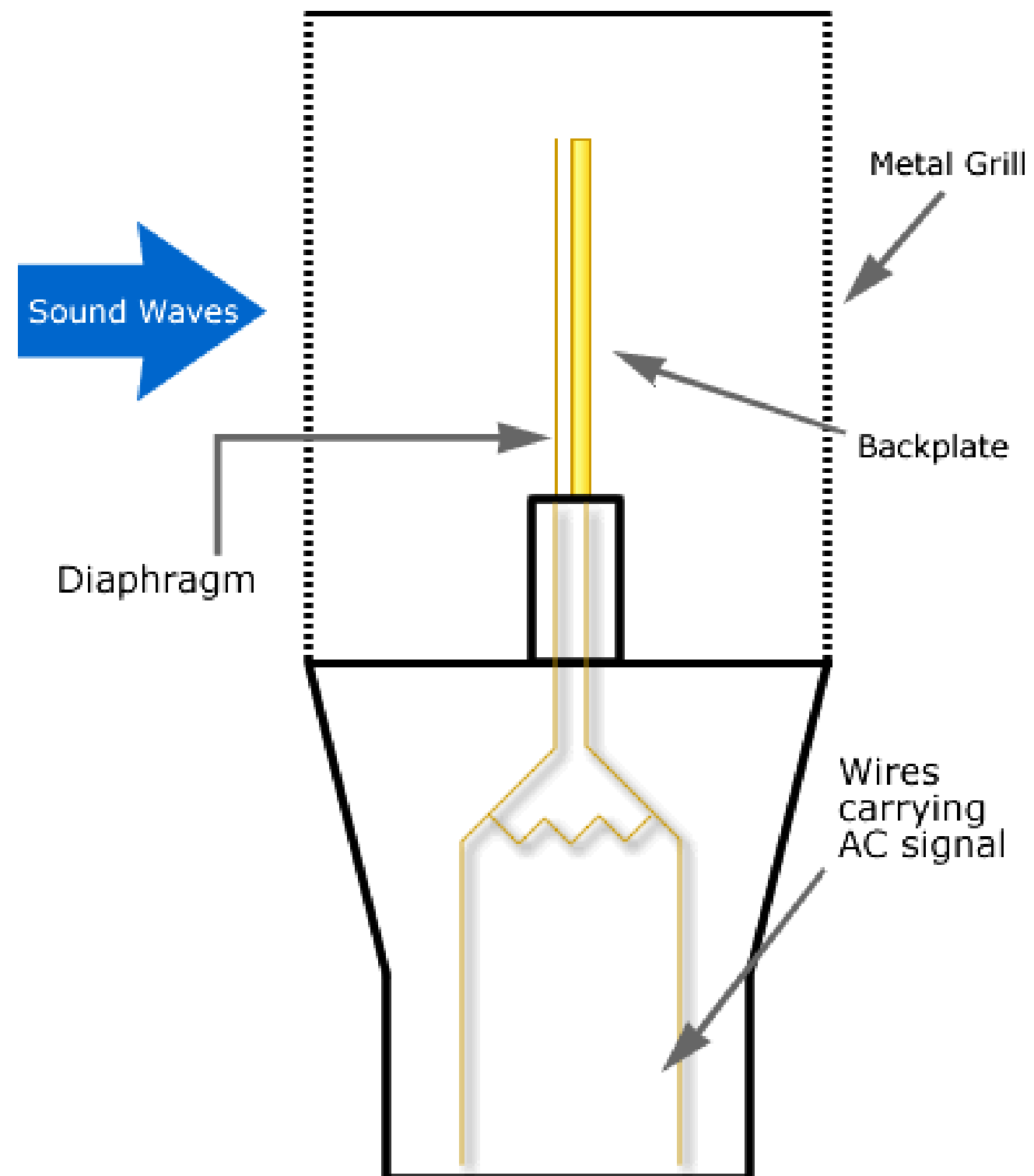
Transducer designs – Condenser mics

- A capacitor (condenser) has two plates with a voltage between them.
- One of these plates is made of very light material and acts as the diaphragm.
- The diaphragm vibrates when struck by sound waves, changing the distance between the two plates and therefore changing the capacitance.
- When the plates are closer together, capacitance increases and a charge current occurs. When the plates are further apart, capacitance decreases and a discharge current occurs.
- A voltage (phantom power +45v) is required across the capacitor for this to work.

Cross-Section of a Typical Condenser Microphone



Cross section of a Condenser mic



Condenser mic facts

- 👁️ Condenser mics have a wide-range frequency response, while dynamic mics typically do not.
- 👁️ Condenser mics' frequency response tends to be uniform, while a dynamic mics' typically is not.
- 👁️ Condenser mics are typically used on acoustic instruments and studio vocals.
- 👁️ Dynamic mics are typically used on guitar amps and drums, and for vocal sound reinforcement.

Lavalier clip-on mics Dynamic, condenser

- An **electret microphone** is a type of condenser microphone, which uses a permanently-charged material.
- An **electret** is a stable material with a permanently-embedded static electric charge
- Due to the high resistance of the material, the charge will not decay for hundreds of years.

Phantom power

- ⑥ Phantom power is usually 48 volts DC (48V) applied to pins 2 and 3 of a condenser mic's XLR connector to power the microphone's electronics. Pin 1 (shield) is the ground for the phantom voltage.
- ⑥ All condenser microphones require power to be supplied through the mic cable via phantom powering.
- ⑥ Phantom power is supplied either by a standalone phantom power supply or by a mixer that has phantom power built in.

Dynamic microphones

- Are recommended for live concerts
- are more suitable for live recordings because of their durability and reliability.
- do not overload or distort as easily as condenser mics.
- generally do not deliver an even response across all frequencies needed in a studio recording.
- amplify certain frequency ranges more than others.

Condenser microphones

- capture a more accurate, "flat" signal.
- are better at reproducing the soft decays and subtle nuances of quieter sounds.
- are most commonly used for recording acoustic instruments and voices.
- introduce less signal noise than a dynamic mic and more accurately reproduce the complete frequency spectrum between 20 Hz and 20,000 Hz.

Lavalier clip-on microphones

- 👁 are inconspicuous and portable.
- 👁 can be easily placed close to the direct source of someone's voice.
- 👁 are commonly used for video shoots and television talk shows.
- 👁 do not work well for recording live concerts or for ambient field recording.
- 👁 can capture only one voice.

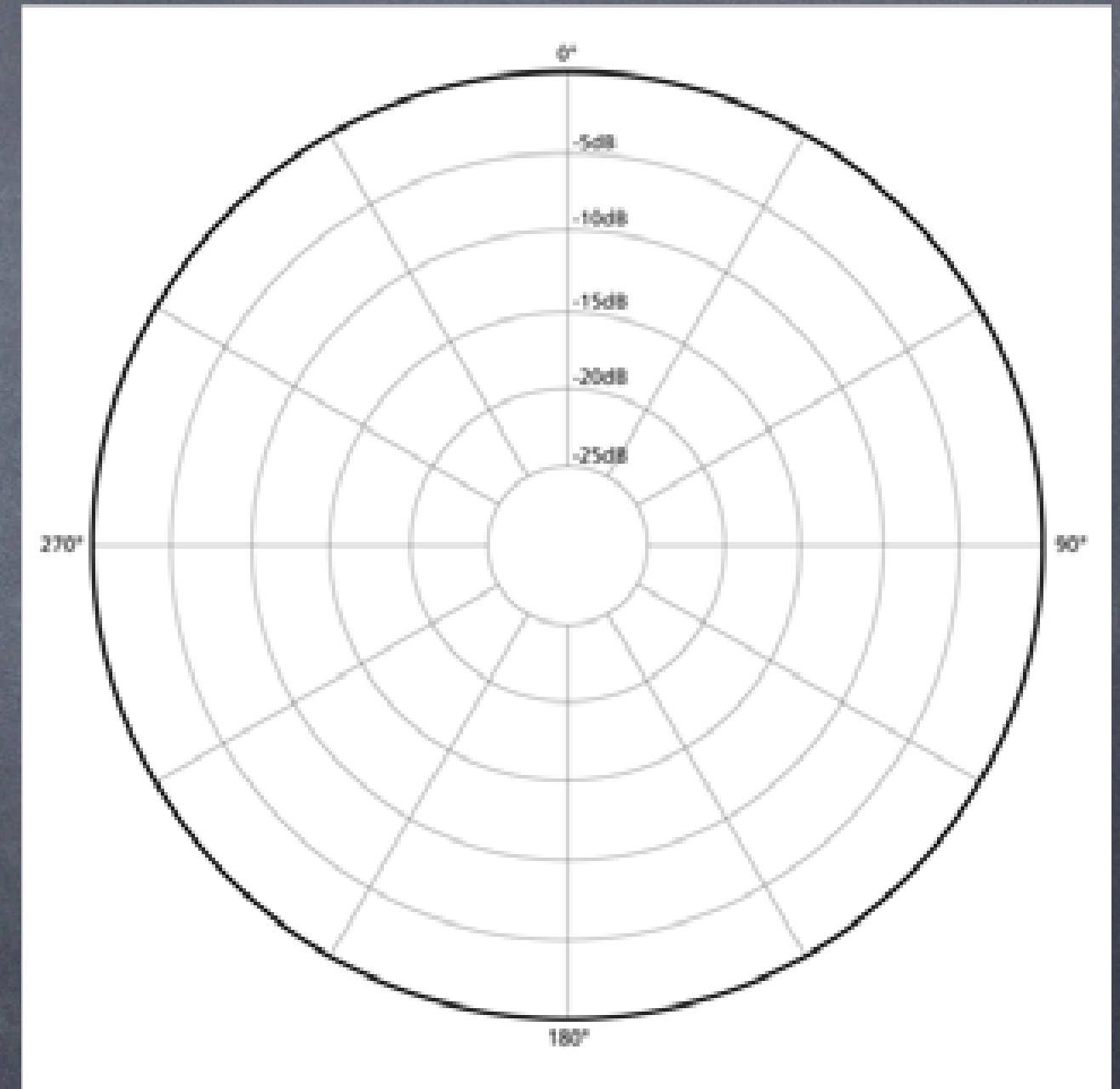
Microphone directionality

Polar Patterns

- 👁️ A polar pattern is a graph of a mic's sensitivity vs. the angle of the incoming sound wave.
- 👁️ The farther from center a point on the graph is, the stronger is the mic signal at that angle.

Omnidirectional:

👁️ Picks up equally in all directions.

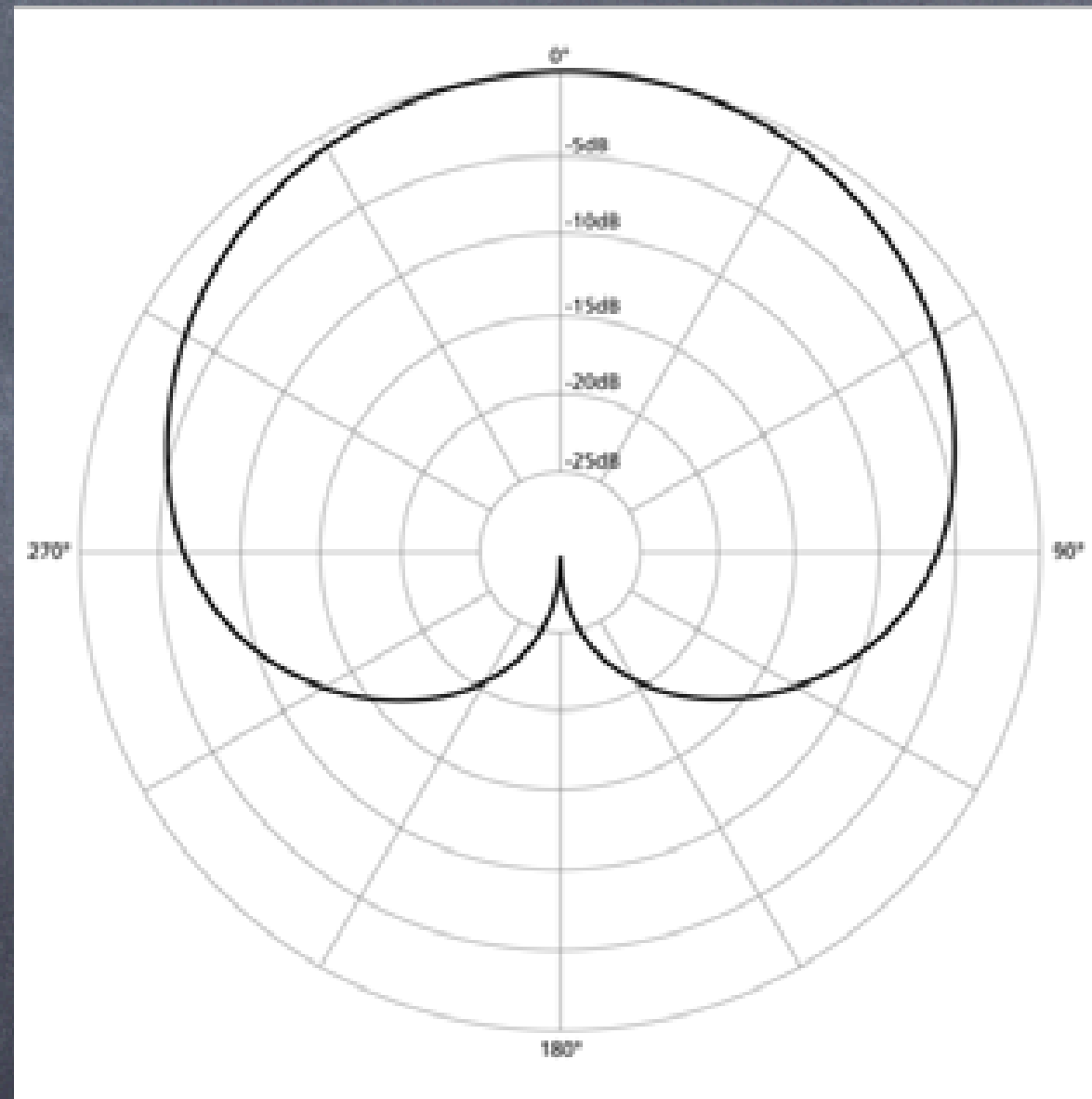


Omnidirectional Traits

- 👁️ All-around pickup
- 👁️ Most pickup of room reverberation
- 👁️ Not much isolation unless you mike close
- 👁️ Low sensitivity to pops (explosive breath sounds)
- 👁️ No up-close bass boost (proximity effect)
- 👁️ Extended low-frequency response in condenser mics.

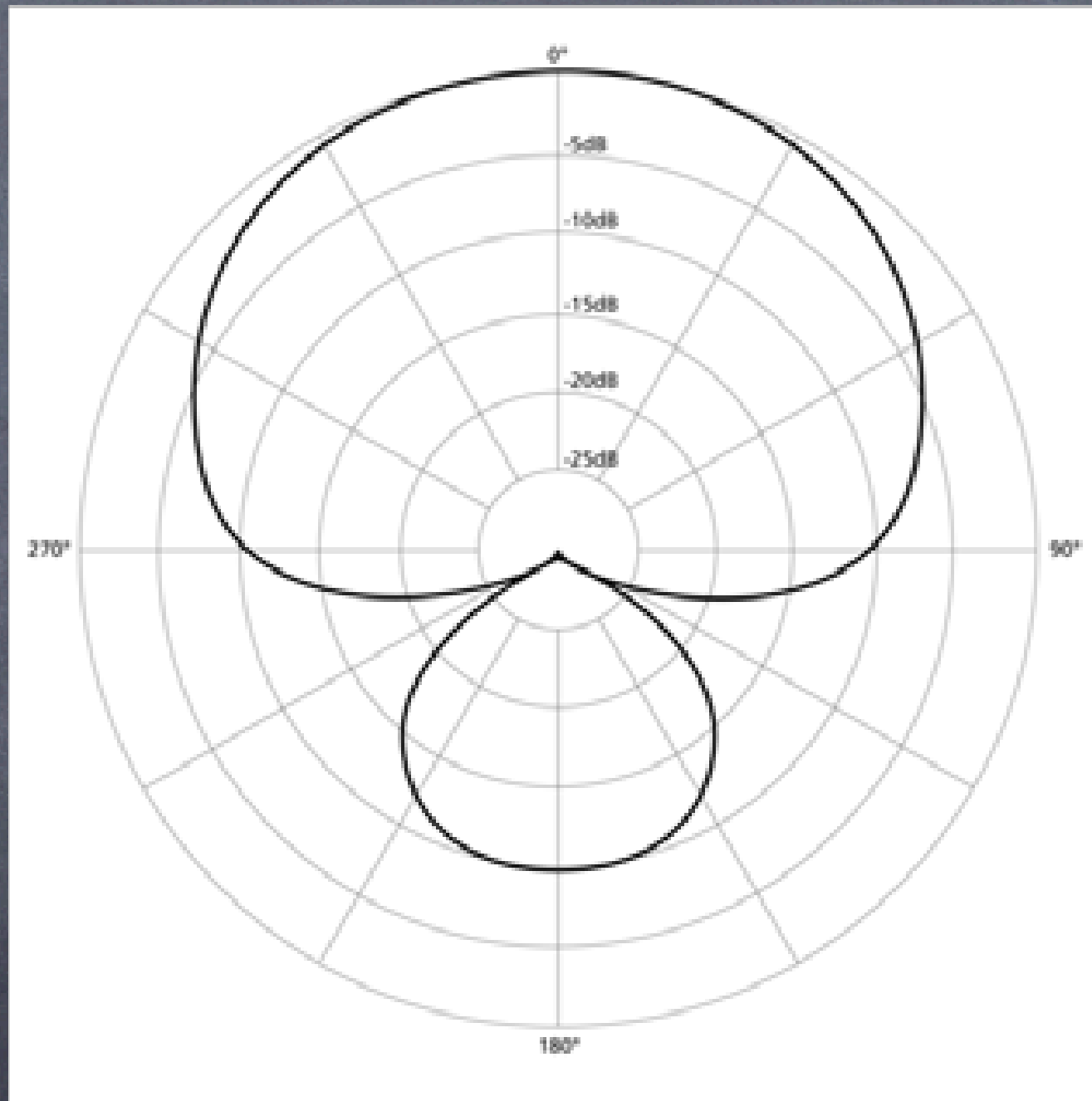
Cardioid:

- “Heart-shaped” pattern that offers maximum rejection (null) at the rear of the microphone. They amplify the bass frequencies of a voice as it gets closer to the microphone, creating the proximity effect.



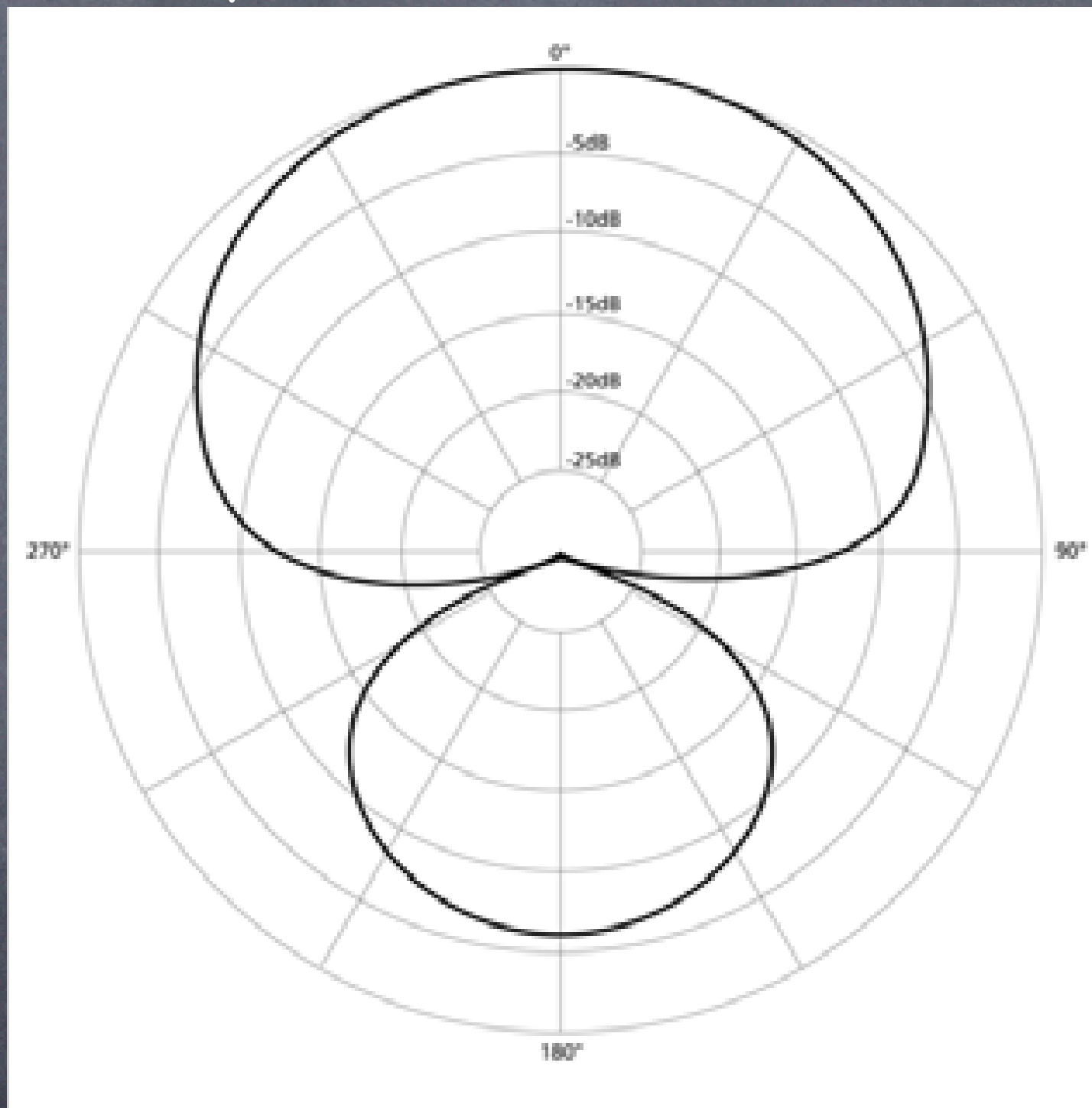
Supercardioid:

- Has a narrower pickup pattern than cardioid, but also has some rear pickup.



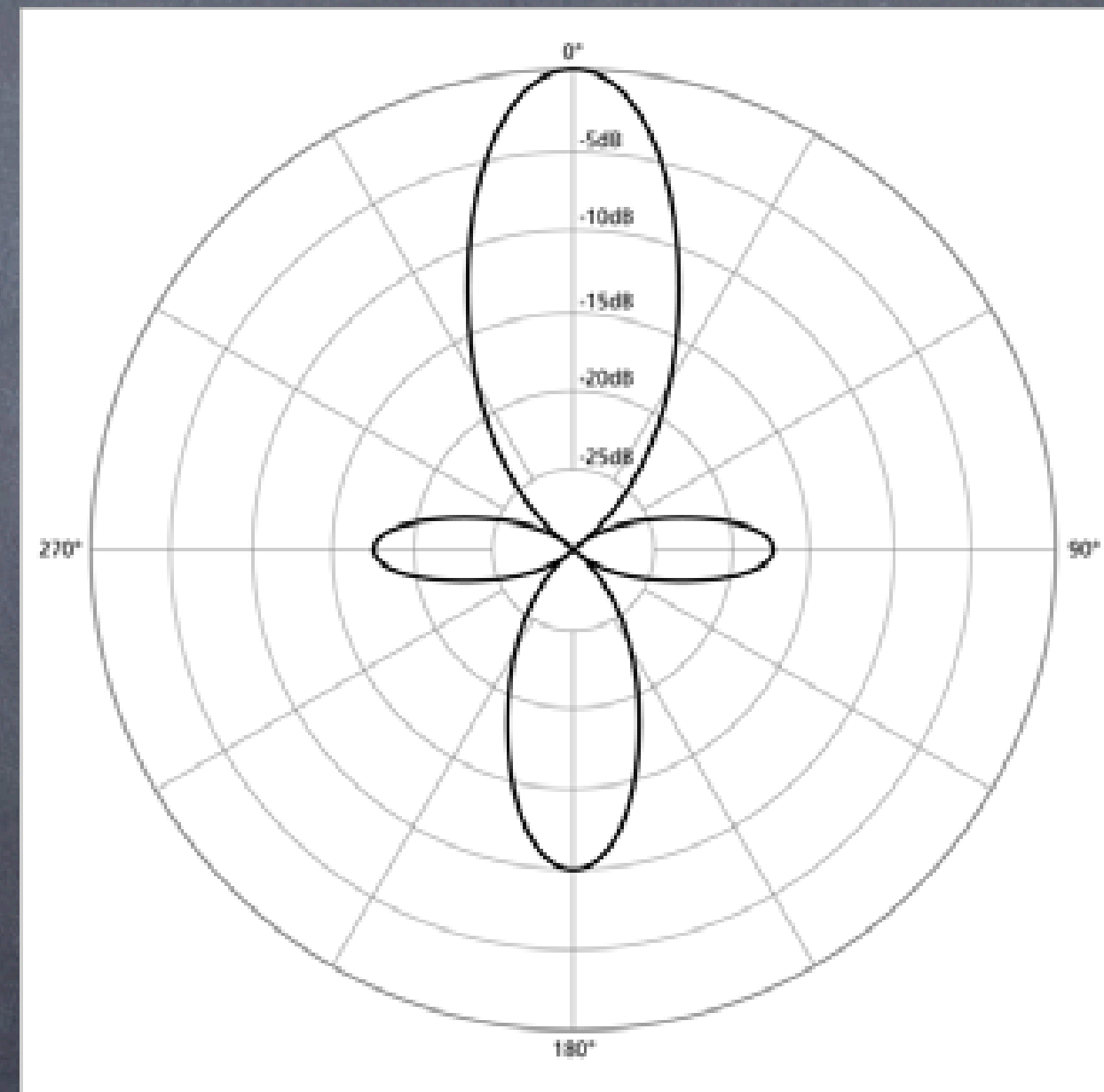
Hypercardioid:

- Has a narrower pickup pattern than supercardioid, but has more rear pickup than supercardioid.



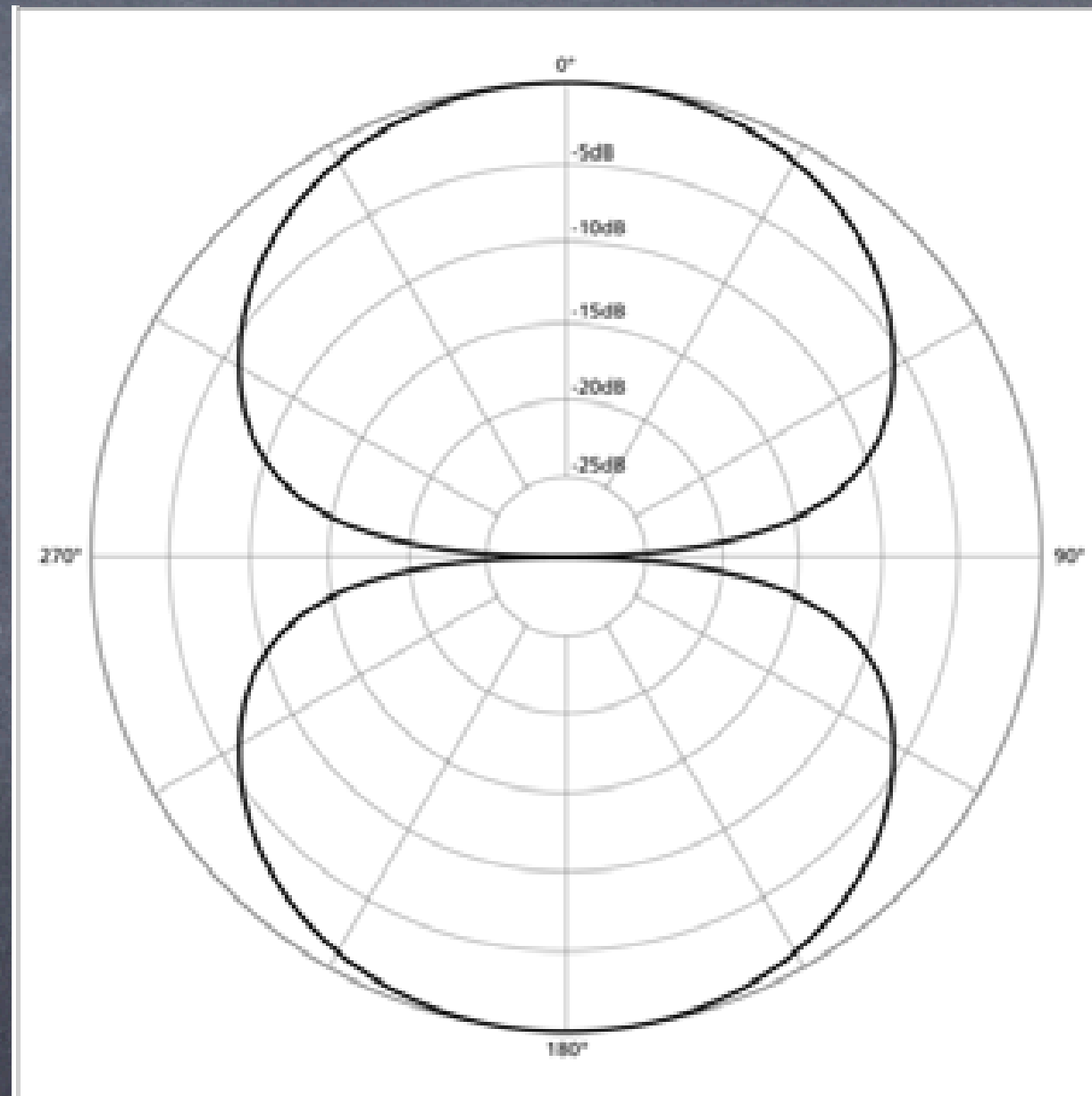
Shotgun microphones

- contain a recording capsule embedded in a long hollow tube.
- The capsule picks up a highly exaggerated hypercardioid pattern.



Bidirectional (figure-eight or cosine):

👁 Picks up mainly in two directions and rejects sound from the sides.



Bidirectional Traits

- ⑥ Front and rear pickup, with side sounds rejected.
- ⑥ Good for across-table interviews or two-part vocal groups.
- ⑥ Also called figure eight.

Polar Patterns Redux



Omni



Bi



Cardioid



Shotgun

How to choose a mic?

- ① Choose an omni if you want the most natural sound and don't mind sound being picked up from all around.
- ② Choose any directional mic if you want to reject sounds from certain directions, or if you want a bass boost such as a vocalist. A cardioid will be a good all-round choice.
- ③ Choose a hypercardioid or shotgun mic if you want a tighter focus than a cardioid.

Mic hints Part 1:

- 👁️ Pay proper attention to the sound at the shoot.
- 👁️ Don't point the microphone and expect perfection.
- 👁️ Don't ever expect issues to be fixed in post. A bad recording will always be a bad recording.
- 👁️ Listen to the room/environment.
- 👁️ What you hear isn't what you get.

Mic hints Part 2:

- 👁️ Mics pick up more reverb than you notice. You can't remove reverb*. Pay attention to where the walls are and how the sound bounces.
- 👁️ You may need to provide sound absorption materials such as sound blankets.
- 👁️ Stay away from walls if you can help it.
- 👁️ Shotgun mics are more directional than hypercardioids though they have similar patterns.

Connectivity XLR

- XLR (3 pin) are recommended because you can have long cable runs with solid connections and very little hum.



- 1 = ground (cable shield) 2 = positive terminal 3 = return terminal

Decibel (Loudness) Comparison Chart

Environmental Noise

- 👁️ Weakest sound heard 0dB
- 👁️ Whisper Quiet Library 30dB
- 👁️ Normal conversation (3-5') 60-70dB
- 👁️ Telephone dial tone 80dB
- 👁️ City Traffic (inside car) 85dB
- 👁️ Train whistle at 500', Truck Traffic 90dB
- 👁️ Subway train at 200' 95dB
- 👁️ Level at which sustained exposure may result in hearing loss 90 - 95dB
- 👁️ Power mower at 3' 107dB
- 👁️ Snowmobile, Motorcycle 100dB
- 👁️ Power saw at 3' 110dB
- 👁️ Sandblasting, Loud Rock Concert 115dB
- 👁️ Pain begins 125dB
- 👁️ Pneumatic riveter at 4' 125dB
- 👁️ Even short term exposure can cause permanent damage - Loudest recommended exposure WITH hearing protection 140dB
- 👁️ Jet engine at 100', Gun Blast 140dB

Mic measurements 1

FREQUENCY RESPONSE 80-18,000Hz

- Frequency response is the measure of any system's spectrum response at the output to a signal of varying frequency (but constant amplitude) at its input.

IMPEDANCE 1.5k Ohms @ 1KHz

- Impedance represents the relationship between voltage and current which a device is capable of accepting or delivering.

*This can become a factor with long cables degrading audio quality

Mic measurements 2

MAX. SPL INPUT LEVEL 110dB

👁 SPL= sound pressure level

SIGNAL TO NOISE RATIO 65dB

👁 Signal-to-noise ratio compares the level of a desired signal (such as music) to the level of background noise. The higher the ratio, the less obtrusive the background noise is.

Mic measurements 3

DYNAMIC RANGE 80dB

- Dynamic range: a ratio (expressed in dBs) of the difference between the softest and the loudest sound that can be produced, reproduced or captured by an audio device.